

CausalGraph: When Causal Reasoning Meets Large Language Models for Intrusion Detection Systems

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Abstract—Modern intrusion detection systems often achieve impressive benchmark accuracy yet fail in real-world deployment, where network behavior and attacker tactics continuously evolve. Under concept drift, decision boundaries learned offline can diverge from operational reality, triggering false-alarm cascades and creating detection blind spots that erode analyst trust. In this paper, we propose CausalGraph-IDS, a causal and language-model-assisted intrusion detection framework that moves beyond purely correlational scoring by explicitly verifying multi-stage attack chains. Additionally, we propose CHAIN-CRC, a unified algorithm that (i) learns a *constrained attack-chain causal graph* guided by knowledge-based priors derived from widely used adversary behavior taxonomies, (ii) computes robustness scores by testing whether alarms persist under feasible counterfactual security interventions, and (iii) applies conformal risk control to enforce operator-defined false-positive budgets with finite-sample guarantees.

Generative models are integrated in strictly assistive roles through three modules: prior induction to distill causal constraints from unstructured threat reports, counterfactual generation to propose realistic and operationally feasible interventions, and causal logic justification to produce human-readable explanations grounded in the learned attack chain. Experiments on two widely used network intrusion detection benchmarks show that CausalGraph-IDS provides robust, explainable, and risk-governed detection, maintaining strong recall at low false-positive rates while delivering actionable causal insights for mitigation.

Index Terms—Intrusion Detection, Causal Reasoning, LLMs, Conformal Risk Control, Generative AI, Concept Drift.

I. INTRODUCTION

Network Intrusion Detection Systems (IDSs) are a cornerstone of modern cybersecurity. However, as attack techniques become more sophisticated and multi-stage, traditional signature- and rule-based approaches struggle to adapt to novel threats and capture long attack chains. Although learning-based IDS (including deep models and Transformers) can improve benchmark performance, they remain largely correlational and sensitive to dataset bias and concept drift.

Large Language Models (LLMs) have recently emerged as powerful tools for security tasks such as CTI parsing and rule synthesis. However, directly using LLMs as IDS decision-makers raises concerns including hallucinations, prompt sensitivity, and the absence of calibration guarantees. Causal reasoning offers a complementary perspective by explicitly modeling relationships between events, attacker actions, and system states, enabling intervention-driven validation of detection hypotheses.

Combining causal reasoning with LLM assistance can therefore enable IDS that detect not only *what* happens, but also *why* it happens and *which interventions* are most effective.

In this paper, we propose **CausalGraph-IDS**, a framework that integrates causal reasoning with LLM assistance to address key limitations of current IDS. Rather than asking only whether an event resembles known attacks, our approach asks a causal question: *Would this alert persist if we intervened to remove its apparent cause?* CausalGraph-IDS builds on an Attack-Chain Structural Model (ACSM), a temporally ordered causal representation of multi-stage attack progressions. We introduce two robustness measures: the Interventional Persistence Index (IPI) and the Counterfactual Detection Score (CDS), which quantify whether detected attack chains remain stable under realistic Security Operations Center (SOC) interventions. To support deployment, we apply Conformal Risk Control (CRC) to provide auditable guarantees on false-positive rates without assuming stationarity. Finally, we use LLMs strictly as assistive components through three modules: Large Language Model-based Prior Induction (LLMPI) to extract causal priors from CTI, SACG to propose realistic counterfactual interventions, and CLJ to generate human-readable causal explanations. For confirmed alerts, the system outputs a Minimal Break-Intervention Set (MBIS), i.e., the smallest set of defensive actions needed to disrupt the inferred attack chain.

The main contributions of this paper are:

- We propose **CausalGraph-IDS**, an intrusion detection framework that grounds alerts in an attack-chain structural causal model (ACSM) rather than purely correlational patterns.
- We introduce a detection principle based on **persistence under intervention**, using the Interventional Persistence Index (IPI) and Counterfactual Detection Score (CDS) to verify whether alerts remain stable under SOC-feasible counterfactual changes.
- We develop CHAIN-CRC, an end-to-end algorithm that couples *constrained causal graph learning*, interventional robustness scoring, and Conformal Risk Control to provide explicit and auditable false-positive guarantees.
- We derive MBIS to map causal insights into minimal, feasible defensive actions and report a drift-aware evaluation protocol (ablation, calibration, and actionability metrics) showing improved robustness, interpretability,

and operational usefulness over strong baselines.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed causal and LLM-assisted framework. Section IV reports experimental results. Finally, Section V concludes the paper and discusses limitations and future work.

II. RELATED WORK

Early learning-based IDS mainly relied on tabular classifiers like Logistic Regression, Support Vector Machines, Random Forests, and gradient-boosted trees trained on flow-level features. These approaches achieve strong pattern-recognition performance, but they are sensitive to dataset bias and concept drift and typically provide limited interpretability or operational guarantees [1], [2]. Most treat detection as a static supervised-learning problem tied to fixed datasets, without explicit reasoning about attack progression or intervention effects.

Recent deep learning and Transformer-based IDS aim to capture non-linear feature interactions and longer-range dependencies across flows and sessions [3]. Graph-based approaches represent entities and communications as dynamic graphs and apply graph neural networks to detect structural anomalies and coordinated attacks [4]. While these methods move toward an attack-chain perspective, they remain fundamentally correlational: alerts are driven by embedding scores without explicit intervention semantics, and calibration is often heuristic. As a result, behavior under counterfactual changes and distribution shifts remains difficult to audit.

LLMs have been adopted as security building blocks for CTI normalization, malware triage, vulnerability analysis, and rule synthesis [5], [6]. For intrusion detection, LLMs have been used to generate or generalize IDS rules and map CTI into ATT&CK-style semantics, but LLM-centric IDS designs can suffer from opaque decision-making and lack formal guarantees on false positives.

In parallel, conformal prediction and Conformal Risk Control (CRC) provide principled tools to enforce deployment-time constraints on risk metrics (e.g., false-positive rate) with minimal distributional assumptions [7]. CRC has begun to appear in anomaly detection and calibration pipelines, but typically without incorporating explicit attack chains or causal interventions. Meanwhile, causal reasoning in cybersecurity has been explored to separate spurious from structural correlations and to reason about defensive actions [8]. However, existing efforts often address causal discovery or counterfactual analysis in isolation, and do not integrate causal structure, intervention-based robustness, and risk-governed alerting into an end-to-end SOC-ready IDS.

CausalGraph-IDS bridges these lines through three key distinctions. First, unlike correlational IDS, it grounds detection in an attack-chain structural causal model and scores alerts by *interventional persistence*. Second, unlike LLM-centric approaches, it restricts LLMs to assistive functions (prior induction, plausible counterfactual proposal, and explanation), while alerting decisions and risk governance remain controlled by the causal model and CRC. Third, by applying CRC at

the chain level, it provides explicit and auditable false-positive guarantees aligned with SOC constraints.

III. CAUSAL REASONING FOR INTRUSION DETECTION SYSTEMS

In this section, we describe the proposed CausalGraph-IDS pipeline, from telemetry to risk-controlled alerts. The framework combines an Attack-Chain Structural Model (ACSM), interventional robustness scores (IPI/CDS), assistive LLM/GAI modules, and Conformal Risk Control (CRC).

A. Structural Causal Models in Cybersecurity

We build on Pearl’s Structural Causal Model (SCM) formalism $M = (X, U, F, G)$, where X denotes observed variables, U exogenous noise variables, G a directed acyclic graph (DAG), and F the structural equations that generate the data [9]. Interventions of the form $\text{do}(X_j = x)$ replace the corresponding mechanism in F , yielding interventional distributions that differ from conditional ones and enabling counterfactual reasoning across alternate ‘worlds.’ In the intrusion-detection setting, we employ temporally windowed representations to capture evolving multi-stage attack chains, and we resort to richer graph variants when conjunctive dependencies arise (e.g., an *Exploit* stage requiring both *Delivery* and a privilege escalation signal). Rather than focusing on effect sizes alone, our detector operationalizes *persistence under intervention* as a robustness criterion: an alert is considered reliable only if the inferred attack evidence remains detectable under a family of targeted, SOC-feasible interventions on the causal representation.

B. System Model

Figure 1 summarizes CausalGraph-IDS, a risk-governed IDS that converts raw telemetry into causally grounded and actionable alerts through six stages. (1) *Telemetry ingestion*: continuous collection of NetFlow, DNS, EDR, and authentication logs provides host/session-level visibility. (2) *Preprocessing and eventization*: logs are segmented into fixed windows, aligned by host and session, and encoded into feature vectors $x \in \mathbb{R}^d$ capturing behavioral signatures. (3) *Attack-Chain Structural Causal Model (ACSM)*: we learn a stage-wise DAG over canonical kill-chain phases $S = \{\text{Recon, Delivery, Exploit, C2, Exfil}\}$. (4) *Intervention-driven robustness*: when suspicious activity is detected, a candidate subchain $G_s \subseteq G$ is selected and tested under SOC-feasible counterfactual interventions. (5) *Risk-controlled decision*: Conformal Risk Control (CRC) enforces an explicit false-positive budget δ with distribution-free guarantees. (6) *Actionable delivery*: confirmed alerts are reported with a natural-language causal rationale and a Minimal Break-Intervention Set (MBIS). LLM modules are strictly assistive: the causal model and CRC govern alerting decisions, while LLM outputs are used only to induce priors, propose plausible interventions, and generate explanations aligned with MITRE ATT&CK semantics [10].

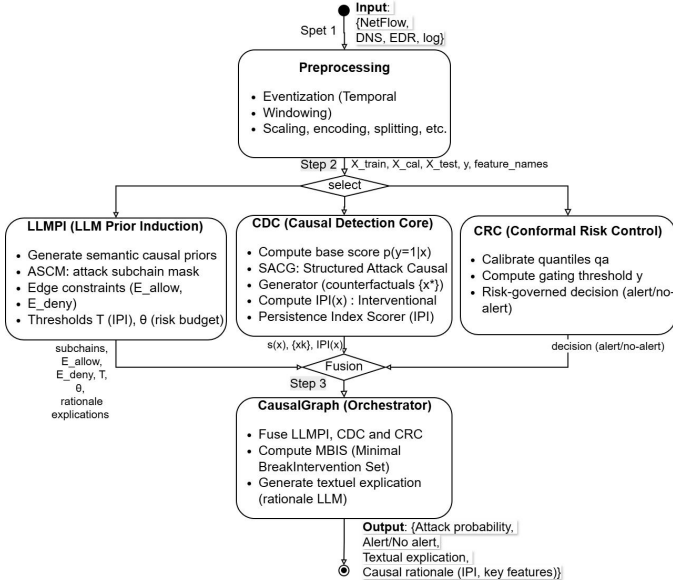


Fig. 1: Overview of the CausalGraph-IDS architecture. The pipeline processes telemetry through stages of ingestion, eventization, causal graph learning (ACSM), intervention-based robustness scoring, risk-controlled decision (CRC), and actionable alert delivery with MBIS and explanations.

C. Interventional Persistence Index (IPI) and Counterfactual Detection Score (CDS)

Given a suspicious subchain $G_s \subseteq G$ and a scalar detector score $\text{score}(G_s, x; \theta) \in [0, 1]$, we evaluate robustness under a catalog $\mathcal{I}$ of *SOC-feasible* interventions. We implement each intervention $\iota \in \mathcal{I}$ as a *plausible feature edit* on the window representation:

$$x^{(\iota)} = \text{Apply}(x, \iota), \quad (1)$$

constrained by operational policies and plausibility invariants (e.g., protocol-port consistency, bounded rate changes).

a) *IPI (binary persistence)*.: We define the *Interventional Persistence Index* as the probability that the subchain remains above a decision threshold τ under interventions:

$$\text{IPI}(G_s, x) = \mathbb{E}_{\iota \sim \mathcal{I}} [\mathbf{1}\{\text{score}(G_s, x^{(\iota)}; \theta) \geq \tau\}]. \quad (2)$$

In practice, we approximate IPI via Monte Carlo sampling with K interventions $\{\iota_k\}_{k=1}^K$:

$$\widehat{\text{IPI}}(G_s, x) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}\{\text{score}(G_s, x^{(\iota_k)}; \theta) \geq \tau\}. \quad (3)$$

High IPI indicates that evidence for the inferred chain persists under realistic SOC actions, while low IPI suggests brittle, intervention-sensitive correlations.

b) *CDS (margin-aware persistence)*.: IPI is binary (above/below τ) and can be insensitive when scores fluctuate near the threshold. We therefore introduce a continuous

Counterfactual Detection Score (CDS) that captures the margin to τ :

$$\text{CDS}(G_s, x) = \mathbb{E}_{\iota \sim \mathcal{I}} \left[\sigma \left(\frac{\text{score}(G_s, x^{(\iota)}; \theta) - \tau}{\beta} \right) \right], \quad (4)$$

where $\sigma(\cdot)$ is the logistic function and $\beta > 0$ is a temperature parameter. We use IPI as a robustness gate ($\text{IPI} \geq \gamma$) and CDS for ranking and for stable calibration when scores concentrate near τ .

D. Risk-Governed Deployment via Conformal Risk Control

We enforce an explicit false-positive budget δ using Conformal Risk Control (CRC) under an exchangeability assumption [11]. Let $y \in \{0, 1\}$ denote benign/attack and define the false-alert risk for a decision $a \in \{\text{benign}, \text{alert}\}$ as:

$$r(a, y) = \mathbf{1}\{a = \text{alert} \wedge y = 0\}. \quad (5)$$

a) *Calibration*.: On a calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$, we compute benign calibration scores

$$c_i = \text{CDS}(G_{s,i}, x_i) \quad \text{for } y_i = 0, \quad (6)$$

and set the operational threshold to the $(1 - \delta)$ quantile of benign scores:

$$t_{\text{neg}}^{(1-\delta)} = \text{Quantile}_{1-\delta}(\{c_i : y_i = 0\}). \quad (7)$$

b) *Deployment rule*.: At deployment time, we issue an alert only if (i) a suspicious subchain exists, (ii) it is robust under interventions, and (iii) it exceeds the CRC operating threshold:

$$\pi_{\delta}(x) = \begin{cases} \text{alert}, & \text{if } G_s \neq \emptyset \wedge \text{IPI}(G_s, x) \geq \gamma \\ & \wedge \text{CDS}(G_s, x) \geq t_{\text{neg}}^{(1-\delta)}, \\ \text{benign}, & \text{otherwise.} \end{cases} \quad (8)$$

Algorithm 1 describes the end-to-end CHAIN-CRC detection loop. The telemetry stream T is first converted into windowed feature vectors X by `EVENTIZEANDENCODE`, where each incoming window $x \in \mathbb{R}^d$ summarizes host/session behavior over a fixed time interval. The CTI corpus $\mathcal{K}$ is processed by `LLM_PRIORINDUCTION` to extract causal constraints: allowed edges E_{allow} , forbidden edges E_{deny} , and a stage partial order π that guide causal discovery. Using these constraints, `LEARNACSM` learns the attack-chain causal graph parameters (W^*, θ^*) (graph structure and scoring parameters). Deployment-time calibration then computes the CRC operating threshold $t_{\text{neg}}^{(1-\delta)}$ on the calibration set $\mathcal{D}_{\text{cal}}$ to satisfy the target false-positive budget δ . For each new window x , `SELECTSUSPICIOUSSUBCHAIN` identifies a candidate subchain G_s , which if non-empty, robustness is evaluated under an intervention catalog $\mathcal{I}$ by estimating the Interventional Persistence Index IPI (fraction of interventions for which the score remains above τ) and the Counterfactual Detection Score CDS (margin-aware persistence). An alert is triggered only if both robustness gates are satisfied, i.e., $\text{IPI} \geq \gamma$ and $\text{CDS} \geq t_{\text{neg}}^{(1-\delta)}$. When triggered, `FINDMINBREAKSET` computes the Minimal Break-Intervention Set (MBIS), and `CLJ_EXPLAIN` produces a

Algorithm 1 CHAIN-CRC: CausalGraph-IDS Detection Pipeline

Require: Telemetry stream T , CTI corpus $\mathcal{K}$, calibration set $\mathcal{D}_{\text{cal}}$, intervention catalog $\mathcal{I}$

Ensure: Risk-controlled alerts with MBIS and explanations

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1:  $X \leftarrow \text{EVENTIZEANDENCODE}(T)$ 
2:  $(E_{\text{allow}}, E_{\text{deny}}, \pi) \leftarrow \text{LLM\_PRIORINDUCTION}(\mathcal{K})$ 
3:  $(W^*, \theta^*) \leftarrow \text{LEARNACSM}(X, E_{\text{allow}}, E_{\text{deny}}, \pi)$ 
4:  $t_{\text{neg}}^{(1-\delta)} \leftarrow \text{CRC\_CALIBRATE}(\mathcal{D}_{\text{cal}}, W^*, \theta^*, \mathcal{I})$ 
5:  $\mathcal{A} \leftarrow \emptyset$   $\triangleright$  Alert set
6: for each new window  $x$  do
7:    $G_s \leftarrow \text{SELECTSUSPICIOUSSUBCHAIN}(x, W^*, \theta^*)$ 
8:   if  $G_s \neq \emptyset$  then
9:      $\text{IPI} \leftarrow \text{ESTIMATEIPI}(G_s, x, \mathcal{I}, \tau, \theta^*)$ 
10:     $\text{CDS} \leftarrow \text{ESTIMATECDS}(G_s, x, \mathcal{I}, \tau, s, \theta^*)$ 
11:    if  $\text{IPI} \geq \gamma$  and  $\text{CDS} \geq t_{\text{neg}}^{(1-\delta)}$  then
12:       $\text{MBIS} \leftarrow \text{FINDMINBREAKSET}(G_s, x, \mathcal{I}, \tau, \theta^*)$ 
13:       $\text{Rationale} \leftarrow \text{CLJ\_EXPLAIN}(G_s, \text{IPI}, \text{MBIS})$ 
14:       $\mathcal{A} \leftarrow \mathcal{A} \cup \{(x, G_s, \text{IPI}, \text{MBIS}, \text{Rationale})\}$ 
15:       $\text{ALERTANALYST}(x, G_s, \text{IPI}, \text{MBIS}, \text{Rationale})$ 
16:    end if
17:  end if
18: end for
19: return  $\mathcal{A}$ 

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human-readable rationale linking G_s , IPI, and MBIS. The algorithm outputs the alert set $\mathcal{A}$, where each alert contains the window identifier, the detected subchain, robustness scores, MBIS, and the explanation.

E. Assistive LLM and Generative Modules

LLMs are used in *non-decision* roles to increase structure, plausibility, and interpretability without the final alert control:

- **LLMPI** (Prior Induction) parses CTI and ATT&CK descriptions [10] to extract edge priors $(E_{\text{allow}}, E_{\text{deny}})$ and a partial order π over stages.
- **SACG** (Counterfactual Generation) proposes *SOC-feasible* interventions (e.g., port normalization, privilege revocation, rate limiting) consistent with policy constraints.
- **CLJ** (Causal Logic Justification) produces concise analyst-facing rationales that link the activated subchain G_s , robustness scores (IPI/CDS), and MBIS to ATT&CK tactics and techniques.

This design follows the emerging trend of using LLMs as security copilots in SOC workflows, while keeping alerting and risk guarantees governed by the causal model and CRC.

F. Qualitative Examples of LLM Assistance

Table I illustrates representative examples of the three LLM-assisted modules: (i) LLMPI extracting ATT&CK-aligned structural priors, (ii) SACG proposing SOC-feasible counterfactual interventions, and (iii) CLJ producing analyst-facing causal explanations.

TABLE I: Representative examples of LLM assistance artifacts (qualitative).

Module	Example output (illustrative)
LLMPI (priors)	Allow edges: (Delivery $\rightarrow$ Exploit), (Exploit $\rightarrow$ C2); Deny edges: (Recon $\rightarrow$ Exfil); Stage order: Recon $\prec$ Delivery $\prec$ Exploit $\prec$ C2 $\prec$ Exfil
SACG (interventions)	do(port $\in$ {80,443}), do(privilege = 0), do(burstiness = baseline), with plausibility checks (protocol/port consistency, bounded rate changes)
CLJ (explanation)	“The alert is robust: IPI = 0.92 and high CDS indicate persistence under port normalization and privilege revocation. MBIS suggests revoking elevated privileges and blocking C2 on uncommon ports, which breaks the Exploit $\rightarrow$ C2 link.”

G. Hypotheses on LLM-Guided Causal Priors

We articulate the role of LLMPI through testable hypotheses:

- **H1** (Structural alignment). LLM-parsed CTI can extract plausible stage orderings and allow/deny edge priors that reduce spurious arcs in the learned ACSM without degrading detection performance.
- **H2** (Operational robustness). ATT&CK-guided constraints improve robustness under strict false-positive budgets, increasing TPR@1%FPR and persistence (IPI/CDS) compared to a purely data-driven SCM.
- **H3** (Consistency and calibration). CLJ explanations aligned with ATT&CK semantics can be provided without harming calibration (low ECE) or overall detection quality (high F1).

These hypotheses translate the intuition behind LLMPI into measurable claims on structural validity, operational reliability, and interpretability.

IV. EXPERIMENTAL EVALUATION

A. Experimental Setup

We evaluate CausalGraph-IDS on two widely used NIDS benchmarks: CICIDS2017, containing enterprise-like traffic and attacks (DoS, brute-force, etc.) [12], and UNSW-NB15, featuring diverse modern attacks (Fuzzers, Backdoors, etc.) generated with IXIA PerfectStorm [13]. For both datasets, we remove redundant attributes, encode categorical fields, and z-score normalize continuous features. We use a stratified 80/20 train/test split.

a) Baselines. We compare CausalGraph-IDS against well-established tabular and deep baselines: Logistic Regression (LR), Random Forest (RF), XGBoost (XGB), and a Transformer-based IDS (TriIDS). All models are tuned on a held-out validation split.

b) Evaluation metrics. We report the Area Under the ROC Curve (AUROC) and Area Under the Precision-Recall Curve (AUPRC) to summarize discrimination across thresholds. We report discrimination, low-false-alarm operating performance, and calibration. In addition, we report two operational metrics tailored to actionable defense.

(i) TPR@1%FPR. This is the true positive rate at the largest decision threshold whose false positive rate does not exceed 1%. Let $TP(\tau)$, $FP(\tau)$, $TN(\tau)$, and $FN(\tau)$ be the confusion-matrix counts at threshold τ :

$$\tau^* = \max \left\{ \tau \mid \frac{FP(\tau)}{FP(\tau) + TN(\tau)} \leq 0.01 \right\}, \quad (9)$$

$$TPR@1\%FPR = \frac{TP(\tau^*)}{TP(\tau^*) + FN(\tau^*)}. \quad (10)$$

(ii) Expected Calibration Error (ECE). We compute ECE with M confidence bins. For bin B_m , let $acc(B_m)$ be empirical accuracy and $conf(B_m)$ the average predicted confidence:

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |acc(B_m) - conf(B_m)|. \quad (11)$$

Lower ECE indicates better probabilistic calibration.

(iii) Security Score. To summarize low-FPR detection quality in a single $[0, 1]$ value, we use:

$$SecurityScore = \frac{1}{2} (F1 + TPR@1\%FPR). \quad (12)$$

This favors detectors that are simultaneously accurate and effective at strict false-alarm constraints.

(iv) Actionability Score. Let $MBIS(x)$ denote the Minimal Break-Intervention Set suggested for alert x (smaller is better), and let K be the size of the intervention library used by the system. We define:

$$Actionability(x) = 1 - \min \left(\frac{|MBIS(x)|}{K}, 1 \right), \quad (13)$$

and report the dataset-level mean actionability over all true-positive alerts. This rewards actionable alerts that can be mitigated with minimal operational disruption.

TABLE II: Ablation study of CausalGraph-IDS on UNSW-NB15 and CICIDS2017.

Variant	F1	Prec.	Rec.	TPR@1%FPR
UNSW-NB15				
Base (no LLM)	0.9924	0.9875	0.9972	0.9951
+ SACG + CRC	0.9942	0.9906	0.9979	0.9975
+ LLMPI + SACG	0.9952	0.9929	0.9975	0.9978
+ LLMPI + CRC	0.9934	0.9951	0.9917	0.9960
Full System	0.9934	0.9952	0.9955	0.9970
CICIDS2017				
Base (no LLM)	0.9582	0.9958	0.9234	0.9314
+ SACG + CRC	0.9582	0.9958	0.9234	0.9315
+ LLMPI + CRC	0.9582	0.9958	0.9234	0.9315
+ LLMPI + SACG	0.9582	0.9958	0.9234	0.9315
Full System	0.9943	0.9980	0.9910	0.9970

On UNSW-NB15, incorporating ATT&CK-aligned priors (LLMPI) and causal-graph reasoning (SACG) improves performance at a strict operating point: TPR@1%FPR increases from 0.9951 (Base) to 0.9978 (+LLMPI+SACG), while maintaining high F1. Adding CRC improves low-false-alarm robustness by calibrating the decision threshold. The full system preserves

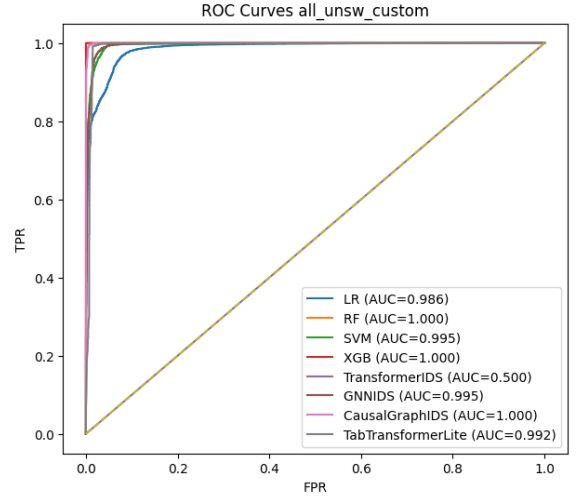


Fig. 2: ROC curve evaluation on UNSW-NB15

a strong precision-recall balance (Prec.=0.9952, Rec.=0.9955) and maintains a high TPR@1%FPR (0.9970). Figure 2 reports ROC curves for both datasets.

On CICIDS2017, several ablations share the same deployment policy (default persistence gate γ and strict CRC calibration at 1% FPR), which can yield identical alert decisions and thus identical metrics up to rounding. The full system additionally uses the complete policy (including tuned γ), explaining the performance jump.

On CICIDS2017, all models achieve near-ceiling AUROC/AUPRC. In this regime, the primary differentiator becomes low-FPR operating behavior and calibration. CausalGraph-IDS achieves the best overall trade-off, with the highest F1 (0.9945), the strongest low-FPR detection (TPR@1%FPR=0.997), and the lowest ECE (0.0029), which is important for risk-aware alerting.

On UNSW-NB15, tree ensembles (RF/XGB) reach near-perfect discrimination. CausalGraph-IDS remains competitive (AUROC=0.9995, AUPRC=0.9996) while providing an operationally meaningful output: causally grounded alert stability, calibrated low-FPR control, and actionable mitigation cues. Compared with LR and TrIDS, it improves the low-FPR operating point substantially (TPR@1%FPR=0.9970 vs 0.7866 and 0.9008) and yields better calibration (ECE=0.0028 vs 0.0245 and 0.0099).

CICIDS2017 shows a *near-saturated* operating regime under our strict 1% FPR constraint. Across several ablation variants, the negative-score tail becomes extremely concentrated, causing CRC to select a threshold close to 1.0. When τ^* is saturated, multiple variants can yield identical alert decisions. In contrast, the full system jointly changes the score distribution through the combination of ATT&CK-informed priors (LLMPI) and counterfactual exposure (SACG), which widens separation at the negative tail and prevents threshold saturation, producing the observed gains.

Our results highlight that *operating-point control* is as

TABLE III: Performance comparison on CICIDS2017 and UNSW-NB15 (some ablation deltas are below 10^{-4}).

Dataset	Model	AUROC	AUPRC	F1	Precision	Recall	TPR@1%FPR	ECE
CICIDS2017	LR	0.9992	0.9982	0.9905	0.9920	0.989	0.992	0.0063
	RF	0.9996	0.9989	0.9837	0.9980	0.970	0.992	0.0666
	XGB	0.9998	0.9995	0.9873	0.9980	0.977	0.997	0.0069
	TrIDS	0.9993	0.9988	0.9894	0.9980	0.981	0.995	0.0075
	CGIDS (ours)	0.9998	0.9996	0.9945	0.9980	0.991	0.997	0.0029
UNSW-NB15	LR	0.9860	0.9880	0.9449	0.9481	0.9418	0.7866	0.0245
	RF	1.0000	1.0000	0.9993	0.9986	1.0000	1.0000	0.0110
	XGB	1.0000	1.0000	0.9998	0.9997	1.0000	1.0000	0.0004
	TrIDS	0.9962	0.9966	0.9770	0.9767	0.9773	0.9008	0.0099
	CGIDS (ours)	0.9995	0.9996	0.9934	0.9952	0.9955	0.9970	0.0028

important as raw discrimination. On UNSW-NB15, LLMPI and SACG improve robustness under strict false-positive constraints by stabilizing the ACSM structure and evaluating persistence under realistic interventions. CRC further enforces a deployment-time risk budget, making the detector behavior auditable for SOC use. IPI and CDS play complementary roles: IPI acts as a robustness *gate* (filtering superficial correlations), while CDS provides a smooth robustness score enabling stable ranking and calibration.

B. Discussion: Operational Interpretation

Our results highlight that *operating-point control* is as important as raw discrimination. On UNSW-NB15, LLMPI and SACG improve robustness under strict false-positive constraints by stabilizing the ACSM structure (ATT&CK-aligned priors) and by evaluating persistence under realistic interventions, which improves TPR@1%FPR. CRC further enforces a deployment-time risk budget, making the detector behavior auditable for SOC use.

The dataset CICIDS2017 is highly separable under the chosen feature representation, and many models reach near-perfect AUROC/AUPRC. At the strict 1% FPR operating point, CRC calibration can saturate the threshold, and when ablations share the same persistence gate, the resulting decision policy can become identical across variants. The Full System uses the complete pipeline and a tuned persistence gate, shifting the operating point and recovering additional true positives while maintaining a very low false-positive rate. Finally, IPI and CDS play complementary roles: IPI acts as a robustness *gate* (filtering superficial correlations), while CDS provides a smooth robustness score enabling stable ranking and calibration, particularly when scores concentrate near the threshold.

V. CONCLUSION

We introduced CausalGraph-IDS, a novel intrusion detection framework that integrates causal reasoning with LLMs to address key limitations in traditional IDS. By replacing purely correlational scoring with causal verification of attack chains, our framework provides a more robust and interpretable approach to intrusion detection. Through the introduction of the **IPI** and CDS, we enable the framework to assess the stability of detected attack chains under feasible security interventions. Additionally, the integration of CRC ensures that the framework

adheres to predefined false-positive budgets, making it suitable for real-world deployment. Our evaluation results show that CausalGraph-IDS outperforms state-of-the-art models in terms of detection performance, interpretability, and risk control.

Future work includes supporting streaming deployment with periodic CRC recalibration, validating the approach on live SOC telemetry, extending CHAIN-CRC to multi-stage Advanced Persistent Threat scenarios, and integrating domain-adapted LLMs for higher-quality CTI priors. We will also study robustness against adaptive attackers targeting the learned causal structure and investigate integration into SIEM/SOAR workflows with analyst-in-the-loop evaluation.

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